Monday, March 24, 2025

LEC24 - Sorting Efficiency and Considerations

Sorting Efficiency

- Time complexity is how the time to sort the list scales with the length of the list n
- Time complexity varies between sorting algorithms
  - Bubble sort → $O(n^2)$
  - Selection sort → $O(n^2)$
  - Insertion sort → $O(n^2)$
  - Quick sort → $O(n \log n)$
  - Merge sort → $O(n \log n)$

Merge Sort

- Recursively splits the list into two halves repeatedly
  - $\log n$ splits
- Halves with 0 or 1 elements are guaranteed to be sorted
  - Base case
- Merge the two halves on the way back
  - $O(n)$ to merge two sorted lists
- Gives overall time complexity of $\log n \; levels \cdot n \; time \; per \; level = O(n \log n)$

Quick Sort

- Efficient sorting algorithm
- Recursively splits the list (by partitioning it) into the part smaller than some value (the pivot) and the part not smaller than that value
- Sort these two parts and recombine the results
- Also $O(n \log n)$, but has a smaller scalar factor, which isnt shown in Big-Oh notation